Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/08/2014
PI Name:	Sibonga, Jean Ph.D.		
Project Title:	Pilot Studies of Radiation Damage in Organ Tissues of Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-4556
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2010	**End Date:**	09/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2014 (previously 9/30/2015) as study was completed, per PI (Ed., 6/3/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	Low doses of radiation have been shown to protect against the biological effects of later exposure to toxic levels of radiation, a phenomenon known as the radioadaptive response. A parent project was funded by the Department of Energy/NASA to Dr. Honglu Wu who will use a mouse model, C57BL/6, to investigate the molecular mechanisms of the phenomenon by systematically identifying the genes that play a role in radioadaptation. To address the specific aims of Dr. Wu's project, his investigative team will expose C57BL/6 mice to Cs-137 gamma rays at a priming low dose followed by a challenging high dose, and investigate the radioadaptive response effect by measuring the endpoints of (1) apoptosis of the cells in the brain, intestine, lung and other organ tissues, and (2) chromosome aberrations in the lymphocytes, bone marrow, and spleen. A total of 200 mice will be used for the DOE project in a period of one year. This animal study, which is now conducted		

Task Description:	at NASA Johnson Space Center, provides a unique opportunity for investigators of other HRP disciplines to conduct a pilot study of radiation damages in the organ tissues that are of interest to the respective discipline. To this aim, Dr. Jean Sibonga is the PI of the sub-project: “Radioadaptive response animals: A pilot study to evaluate effects in bone tissue.” The pilot study will involve a preliminary experiment with 12 mice and a comprehensive experiment with 36 mice. In the preliminary experiment, the mice will be exposed to a low dose of 0.05 Gy, followed by a high dose of 6 Gy 24 hr later. The mice will be sacrificed three days after the high dose exposure. More time points in both the time interval between the priming and the challenging doses, and the time to sacrifice the animal after the challenging dose will be added in the comprehensive experiment. The Bone analysis will conduct bone histomorphometry and microCT analysis of tibia and lumbar vertebrae (L1 and L2).
Rationale for HRP Directed Research:	This cross-discipline study cannot be conducted in human beings. The funded study, awarded to an investigator in the Radiation element, provided a unique opportunity and specimen resource for discipline-specific evaluations by HHC scientists.
Research Impact/Earth Benefits:	
Task Progress:	Radiation-induced death of marrow cells is associated with 1) a transient increase in bone formation due, at least in part, to activation of bone lining cells, and 2) an increase in bone resorption due to increased osteoclast perimeter. Bone marrow transplantation is effective in mitigating the detrimental effects of acute exposure to high dose whole body radiation on bone turnover. High dose whole-body gamma radiation results in a dramatic bone turnover response in which both formation and resorption are increased. The increased turnover is temporally associated with bone marrow failure and may result from disturbed regulation of osteoblast and osteoclast differentiation by hematopoietic cells. Bone marrow transplantation, by restoring normal bone marrow function, is effective in restoring normal bone turnover following lethal irradiation.
Bibliography Type:	Description: (Last Updated: 05/24/2021)
Articles in Peer-reviewed Journals	Turner RT, Iwaniec UT, Wong CP, Lindenmaier LB, Wagner LA, Branscum AJ, Menn SA, Taylor J, Zhang Y, Wu H, Sibonga JD. "Acute exposure to high dose gamma-radiation results in transient activation of bone lining cells." Bone. 2013 Nov;57(1):164-73. [Epub 2013 Aug 13] http://dx.doi.org/10.1016/j.bone.2013.08.002 ; PubMed PMID: 23954507 ; PubMed Central PMCID: PMC4042434 , Nov-2013